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Subject Code: BMC051

Roll No:

MCA

(SEM. IV) THEORY EXAMINATION 2025-26

MOBILE COMPUTING

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x7 = 14

(a)	Define mobile computing and mention any two key characteristics of a mobile computing system.	CO1	K1
(b)	Explain the cellular concept and state its importance in wireless telephony.	CO1	K2
(c)	List any four MAC-related issues in Wireless LANs.	CO2	K1
(d)	What is Mobile IP? State its role in supporting user mobility.	CO2	K2
(e)	Define data replication in mobile computing and mention one advantage of replication.	CO3	K2
(f)	Identify two major security issues in mobile computing and briefly state their impact.	CO4	K4
(g)	Name any two ad hoc routing protocols and classify them as proactive or reactive.	CO5	K4

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

(a)	Explain the GSM air interface and channel structure. Also, describe the role of HLR and VLR in location management.	CO1	K2
(b)	Analyze the problems of TCP over wireless networks. Suggest suitable techniques to improve TCP performance in a wireless environment.	CO2	K4
(c)	Discuss the major data management issues in mobile computing with reference to caching, data replication, and disconnected operations.	CO3	K2
(d)	Analyze the security and fault-tolerance requirements of mobile agent computing. Give suitable examples.	CO4	K4
(e)	Compare GSR, DSDV, DSR, AODV, and TORA routing protocols based on route discovery, maintenance, overhead, and suitability for mobile ad hoc networks.	CO5	K5

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	Describe location management in cellular systems. Explain HLR-VLR-based management, hierarchical location management, and handoff handling with a neat diagram.	CO1	K2
(b)	Explain channel allocation in cellular systems. Discuss fixed, dynamic, and hybrid channel allocation along with CDMA and GPRS support in mobile networks.	CO1	K2

4. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain the IEEE 802.11 architecture and MAC mechanism. Discuss hidden terminal and exposed terminal problems with suitable solutions.	CO2	K4
(b)	Describe WAP architecture, WAP protocol stack, application environment, and its major applications in mobile computing.	CO2	K2

5. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain data replication for mobile computers. Analyze the need for consistency management and adaptive clustering in mobile wireless networks.	CO3	K2
(b)	Discuss file system support for mobile computing. Explain disconnected operations, reintegration, hoarding, and conflict resolution.	CO3	K2

6. Attempt any one part of the following:

7 x 1 = 7

(a)	Analyze different types of security issues in a mobile computing environment. Discuss authentication, confidentiality, integrity, availability, and secure transaction support.	CO4	K4
(b)	Explain transaction processing in a mobile computing environment. Discuss the challenges caused by mobility, disconnection, weak connectivity, and failure recovery.	CO4	K4

7. Attempt any one part of the following:

7 x 1 = 7

(a)	Evaluate DSDV, DSR, AODV, and TORA routing protocols for ad hoc networks. Which protocol is more suitable for highly dynamic networks? Justify your answer.	CO5	K5
(b)	Explain localization, MAC issues, QoS requirements, and important applications of ad hoc networks.	CO5	K4